Rigil Kent O. Biaco 2526 COMP 009 Object Oriented Programming

1. What is Java programming language? Discuss its key features and advantages.

Java was originally designed by James Gosling at Sun Microsystems. Released in May 1995 to develop a language for digital devices like set-top boxes and televisions. Now owned by Oracle Corporation. Java uses object-oriented programming, which means that you can organize code using "classes" and "objects" to make programs easier to manage. Java also has built-in security, can handle multiple tasks at the same time (multithreading). Java's ease of use, accessibility, versatility, and reliability makes it a widely used programming language.

2. Explain the concept of platform independence in Java. How does Java achieve platform independence?

In Java, Platform independence means you can write your program once and run it on different computer without changing the code. Instead of turning your program directly into a format that only one type of computer understands, Java first converts it into something called bytecode. The bytecode runs on the Java Virtual Machine (JVM), which acts like a translator between your program and the computer's operating system. As long as the computer has a JVM installed, it can run on the same Java program whether its Windows, macOS, or Linux. This is why Java is known for "write once, run anywhere."

3. Discuss the main components of a Java program and their roles.

A Java program is made up of several important parts that work together to make it run. First, everything is written inside a class, that acts like a container or blueprint of the program. Inside the class is the main() method, which is the starting point where the program begins to execute. Java variables store data (like numbers or text), along with data types to define what kind of data is being stored. Java also have import tools that are useful for putting together repeatable codes and increase its ease of use.

4. What are the differences between JDK, JRE, and JVM in Java?

- JDK (Java Development Kit)
  → includes the JRE, JVM, and development tools that serves as the complete package to help you write and compile Java Programs.

- JVM (Java Virtual Machine)
  → The part that runs Java Programs. This converts the compiled Java code into machine code so you can understand it.

- JRE (Java Runtime Environment)
  → Includes the Java Virtual Machine plus the necessary libraries and files needed to run Java applications but does not include the specific tools to make programs

5. Explain the difference between interpreted and compiled. Is Java an interpreted or compiled language?

Java is actually both a compiled and interpreted language, since the Java compiler turns the code into bytecode, then the Java Virtual Machine interprets or compiles that bytecode so it can run on any computer. In simpler terms, the JVM converts the bytecode into a program that can adjust itself to the environment of the chosen medium (i.e. smartphones, pc, televisions). That's why Java is such an reliable and dynamic programming language also known as a "hybrid" language.